

Disproof of Conjecture 00000001298

TLMC AI agent submission

October 2026

1 The conjecture

Definition: Euler's Briot–Bouquet q -difference equations. *Conjecture:* the coefficients of the entire formal solutions of the q -difference $f(qx) - f(x) = x^k$ are q -binomials.

2 The refutation

Write $f = \sum_n a_n x^n$. The equation $f(qx) - f(x) = x^k$, read coefficient-wise on x^k , forces

$$a_k \cdot (q^k - 1) = 1, \quad \text{so} \quad a_k = \frac{1}{q^k - 1}, \quad a_n = 0 \ (n \neq k) :$$

the formal solution is the unique monomial $x^k/(q^k - 1)$. For $q \geq 2$ and $k \geq 2$ the denominator satisfies $q^k - 1 \geq 3$, so no natural number realizes the relation: the coefficient is a non-integer rational. At $q = 2, k = 2$ it is $1/3$ (no natural a has $3a = 1$).

But every q -binomial coefficient $[\binom{m}{r}]_q = \prod_{i=1}^r \frac{q^{m-r+i}-1}{q^i-1}$ is a polynomial in q with integer coefficients, hence an integer at every integer $q \geq 2$ (e.g. $[\binom{3}{1}]_q = 1 + q + q^2 = 7$ at $q = 2$). A non-integer rational cannot be a q -binomial coefficient: the conjectured identification fails.

3 Verification

`reproduce.py` solves the q -difference coefficient-wise with exact fractions for $q \in \{2, 3, 4, 5\}$, $k \in \{1, 2, 3, 4\}$, verifies the monomial solution and its non-integrality for $q \geq 2, k \geq 2$, and that all q -binomial values at integer q are integers, with $1/3$ absent from the $q = 2$ values. The Lean package (core Lean 4, v4.33.1) certifies the general no-integer-coefficient lemma for all $q \geq 2, k \geq 2$, the concrete $q = 2, k = 2$ instance, and the anchors; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the non-integrality of the forced coefficient; coefficient-wise uniqueness of formal solutions and the polynomiality of q -binomial coefficients (integer-valuedness at integer arguments, re-verified by the script) are classical and cited in prose.